DARKWATCH XAI OPERATIONAL PLATFORM — MASTER ENGINEERING SPECIFICATION

GHOST PROTOCOL

Autonomous Cyber Threat Observation, OSINT Intelligence, Graph Memory, and XAI System

Document Version: 2.0.0 (Master Release) | Target: Antigravity Core Engineer | Scope: Full Pipeline & Agent Architecture

> **MASTER ENGINEERING SPECIFICATION NOTICE:** This document defines the complete, production-ready system design for **GHOST PROTOCOL**. It governs the end-to-end implementation of autonomous threat observation, entity extraction, identity disambiguation, temporal graph correlation, natural-language investigation, two-tier XAI, and SOC workstation UI. All existing DarkWatch ML models (HackerSignal DistilBERT, XGBoost, Temporal Autoencoder v2, HM-TIF) and Whonix isolation boundaries are **strictly frozen**.

1. SYSTEM IDENTITY & MISSION PHILOSOPHY

GHOST PROTOCOL is an autonomous, fail-closed Cyber Threat Intelligence (CTI) observation, correlation, and explanation system built upon the DarkWatch XAI infrastructure. Unlike primitive keyword tools that rely solely on explicit CVE mentions, GHOST PROTOCOL understands complex, unstructured threat discourse—such as initial access broker listings, stealer log dumps, ransomware extortion claims, and threat actor chatter across darkweb hidden services, public Telegram channels, and paste repositories.

The system operates on four fundamental operational tenets: **(1) Understand** unstructured technical and intent-based context; **(2) Remember** entity lifecycles, first-seen/last-seen timestamps, and state shifts over time; **(3) Correlate** threat actors, aliases, infrastructure, and target organizations into a temporal heterogeneous knowledge graph; and **(4) Investigate** via evidence-grounded natural-language querying and two-tier Explainable AI.

2. SUBSYSTEM & AGENT ARCHITECTURE

Subsystem / Agent Module	Responsibility & Execution Boundary	Primary Input / Output
Discovery & Validation Engine	Queries Ahmia, Daunt.link, deepdarkCTI feeds. Validates reachability via Whonix SOCKS5.	In: Seeds / Out: Validated source queue in darkwatch.db
Hybrid Dual-Engine Observer	Continuous JS-disabled httpx polling + Playwright browser context for Dread 2-step verification.	In: Source URL / Out: Raw HTML + Session state hashes
Preprocessing & Sanitizer	Trafilatura clean body parsing, PII/Crypto/Onion token masking, 128-perm MinHash LSH deduplication.	In: Raw HTML / Out: Sanitized text + MinHash signatures
Frozen DarkWatch ML Brain	HackerSignal DistilBERT (text), XGBoost (network), Temporal Autoencoder v2 (temporal), HM-TIF Fusion.	In: Multi-modal triplet / Out: Fused Risk Score (0-1)

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TIJERE CTI & Graph Memory	Joint entity-relation extraction, Composite Entity Resolution (Jaro-Winkler + Cosine), SQLite Graph.	In: Text + Predictions / Out: Resolved Entities & Graph Edges
STIX 2.1 & MITRE ATT&CK; Engine	Generates 7 SDO types, SCOs, SROs into STIX Bundles; maps Enterprise ATT&CK; T-codes (T1190, T1486).	In: CTI Entities / Out: OASIS STIX 2.1 JSON Bundles
Two-Tier XAI Generator	Tier 1 DistilBERT token attention highlights + Tier 2 SHAP feature attributions (triggered at Risk ≥ 0.70).	In: Model states / Out: Visual explanation payloads
Transient Sherlock OSINT Engine	Async surface-web handle lookup across public platforms. Non-persisted transient UI results.	In: Username handle / Out: Transient OSINT profile links
Analyst SOC Workstation	Interactive web dashboard, Explained Alert Cards, STIX export buttons, Natural-Language Search API.	In: CTI Queries / Out: Visual alert cards & Graph views

3. SOURCE DISCOVERY, HYBRID OBSERVATION & SECURITY BOUNDARY

Whonix Fail-Closed Isolation: All external network communications strictly pass through the Whonix-Gateway SOCKS5 proxy (socks5h://10.152.152.10:9050) enforcing remote DNS resolution. A pre-flight socket verification check executes prior to every request; if the proxy is unreachable, the collector halts immediately to prevent IP leaks.

Source Discovery Mechanism: GHOST discovers public CTI targets via three automated channels: (1) Ingesting signed onion mirrors from **Daunt.link** for Dread; (2) Parsing CTI feeds from **deepdarkCTI**; (3) Telegram public web previews (e.g., t.me/s/SliceForLifeee) without login; and (4) Recursive link hopping extracting .onion and t.me/ URLs from observed content.

Hybrid Dual-Engine Observer: Simple static content is collected via JS-disabled httpx.AsyncClient workers. For session-gated platforms like Dread requiring 2-step verification, an authorized Playwright SOCKS5 browser context completes the interactive login flow once. The resulting session cookies are exported to an *In-Memory Volatile Vault*. Raw secrets are omitted from storage; only **SHA-256 cookie hashes** are logged in darkwatch.db.

4. ENTITY RESOLUTION & TEMPORAL KNOWLEDGE GRAPH

GHOST PROTOCOL extracts 6 primary entity categories: *Company/Organization*, *Threat Actor/Handle*, *Malware/Ransomware*, *Infrastructure (IP, Domain, Wallet, Hash)*, *Stolen Data/Access Claim*, and *Campaign*.

To disambiguate aliases and recognize recurring threat actors without blind merging, the **Entity Resolution Engine** calculates a composite similarity score between a newly observed handle \(E_{\text{new}}\) and canonical entity \(E_{\text{canon}}\):

> **Composite Entity Resolution Score:**
> $S_{comp} = 0.40 \cdot JaroWinkler(E_{new}, E_{canon}) + 0.35 \cdot CosineSim(V_{new}, V_{canon}) + 0.25 \cdot GraphJaccard(N_{new}, N_{canon})$
>
> • **$S_{comp} \geq 0.85$:** Auto-merge into canonical entity; update last_seen_utc and co-occurrence count.
> • **$0.65 \leq S_{comp} < 0.85$:** Create candidate alias edge in entity_aliases for analyst adjudication.
> • **$S_{comp} < 0.65$:** Instantiate new canonical entity record.

5. DATABASE SCHEMA & DATA PERSISTENCE (DARKWATCH.DB)

Table Name	Key Fields & Constraints	Description & Governance

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sources_registry	source_id (PK), canonical_url, source_type, validation_status, raw_html_sha256, clean_text_sha256, minhash_sig	Tracks target sites, validation states (200/503/401), content hashes, and discovery provenance.
canonical_entities	entity_id (PK), canonical_name, entity_type, first_seen_utc, last_seen_utc, total_observations	Primary registry for actors, companies, malware, infrastructure, and access claims.
entity_aliases	alias_id (PK), entity_id (FK), alias_string, match_score, evidence_snippet, status	Stores candidate and confirmed threat actor handle aliases with resolution confidence.
graph_relationships	edge_id (PK), source_entity_id, target_entity_id, relationship_type, confidence_score, provenance_id	Temporal graph edges (e.g. OPERATES, USES, TARGETS, EVIDENCE_FOR) with evidence links.
gold_records	record_id (PK), raw_html_sha256, clean_text_sha256, sanitized_text, stix_bundle_json, mitre_codes, risk_score, xai_payload_json	Complete immutable observation record containing ML scores, STIX bundles, XAI, and provenance.

6. WORKED OPERATIONAL SCENARIOS (EXAMPLES A - F)

Scenario Example	Observed Text / Event Context	GHOST PROTOCOL Internal Processing & Output
Example A: RCE Post with CVE	*'Selling admin credentials & RCE exploit for enterprise server. CVE-2023-4966. Contact BTC 1A1z...'*	Extracts CVE-2023-4966 & BTC wallet. DistilBERT: rce (0.94). STIX 2.1 SDOs generated. MITRE: T1190. Risk: 0.88. Tier 1 token highlights active.
Example B: Stealer Log Dump	*'RedLogs Cloud drop: 15,000 logs containing logins for acme.com, banking portals. Download link...'*	No CVE present. Pre-filter detects breach terms. DistilBERT: auth_access (0.91). Target: Acme Corp. Creates AccessClaim entity. Risk: 0.82.
Example C: Handle Tracking	*'User ShadowFox offering access on Dread, later seen on Telegram as ShadowFox_Official.'*	Entity Resolver matches handles (S_{comp} = 0.78). Creates candidate alias edge. On-demand Sherlock trigger queries public platforms without DB storage.
Example D: Infrastructure Shift	*'Ransomware portal onion1.onion offline. New mirror onion2.onion published with same LockBit victim list.'*	MinHash LSH detects near-duplicate content (J = 0.88). Flags NEAR_DUPLICATE_MIRROR. Links new onion address to existing Campaign entity.
Example E: No-CVE Breach Claim	*'Selling 50GB internal database of GlobalLogistics Ltd. Price $10k Monero. Seller: DarkRider.'*	Extracts Company: GlobalLogistics, Volume: 50GB, Crypto: Monero. DistilBERT: info_disclosure (0.89). Maps to T1486. Risk: 0.85.
Example F: Source Outage	*'Target forum returns HTTP 503 DDoS wall, later recovers and changes domain.'*	Collector logs AUTOMATED_ACCESS_BLOCKED. HM-TIF fusion maintains risk scoring via network/temporal streams. Updates source registry upon recovery.

7. IMPLEMENTATION READINESS & ANTIGRAVITY EXECUTION ORDER

Implementation Status Matrix:

• **ALREADY IMPLEMENTED:** Whonix SOCKS5 Proxy Connection, Trafilatura Clean Text Parsing, PII/Crypto/Onion Token Masking, MinHash LSH Engine, HackerSignal DistilBERT Model, Multiclass XGBoost Model, Temporal Autoencoder v2, HM-TIF Cross-Attention Fusion, TIJERE Joint Extractor, STIXGenerator, MITREMapper, Two-Tier

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1. **Step 1: Database Schema Expansion (darkwatch/storage/annotation_store.py)**
   Add sources_registry, canonical_entities, entity_aliases, and graph_relationships tables to darkwatch.db.
2. **Step 2: Source Discovery & Registry Manager (darkwatch/discovery/search_discoverer.py)**
   Implement Daunt.link mirror resolver for Dread, deepdarkCTI feed parser, and Ahmia search index fetcher.
3. **Step 3: Public Telegram & Session Observer (darkwatch/observer/telegram_public_observer.py)**
   Implement no-login HTML preview observer for t.me/s/SliceForLifeee and Playwright SOCKS5 manager for Dread 2-step verification.
4. **Step 4: Composite Entity Resolution Engine (darkwatch/resolution/entity_resolver.py)**
   Implement Jaro-Winkler + vector similarity + graph Jaccard composite scoring for candidate alias disambiguation.
5. **Step 5: Transient Sherlock OSINT Wrapper (darkwatch/investigation/sherlock_wrapper.py)**
   Build on-demand, non-persisted handle lookup tool querying public platforms.
6. **Step 6: Ingestion Bridge Wiring (darkwatch/pipeline/lab_bridge.py)**
   Chain Stage 1 observations directly into frozen ML, STIX 2.1, MITRE ATT&CK;, and XAI pipeline.
7. **Step 7: SOC Analyst Workstation UI (darkwatch/dashboard/app.py)**
   Build web workstation with Explained Alert Cards, Graph View, STIX Exports, and Search API.
8. **Step 8: Master System Integration Runner (run_ghost_protocol.py)**
   Assemble single-command end-to-end execution runner and run verification test suite.